

Quiz 2 Solution

1. Distance Calculation

We use the **Euclidean Distance** formula to calculate the distance between New Customer F (4, 3) and all existing customers (x, y):

$$d = \sqrt{\{(x_2 - x_1)^2 + (y_2 - y_1)^2\}}$$

Customer	Features (x, y)	Calculation	Distance	Category
A	(2, 1)	$\sqrt{\{(4 - 2)^2 + (3 - 1)^2\}}$ $= \sqrt{\{4 + 4\}}$	2.83	L
B	(3, 2)	$\sqrt{\{(4 - 3)^2 + (3 - 2)^2\}}$ $= \sqrt{\{1 + 1\}}$	1.41	L
C	(6, 5)	$\sqrt{\{(4 - 6)^2 + (3 - 5)^2\}}$ $= \sqrt{\{4 + 4\}}$	2.83	H
D	(7, 6)	$\sqrt{\{(4 - 7)^2 + (3 - 6)^2\}}$ $= \sqrt{\{9 + 9\}}$	4.24	H
E	(5, 4)	$\sqrt{\{(4 - 5)^2 + (3 - 4)^2\}}$ $= \sqrt{\{1 + 1\}}$	1.41	H

2. Identify K=3 Nearest Neighbors

Sorting the distances in ascending order:

1. **Customer B** (Distance: 1.41) → **Category: L**

2. **Customer E** (Distance: 1.41) → **Category: H**

3. **Customer A** (Distance: 2.83) → **Category: L** (*Tie with C, picked A by order*)

3. Majority Voting

- **Category L:** 2 Votes (B, A)
- **Category H:** 1 Vote (E)

Final Prediction: Since Category **L** has the majority, **Customer F** is classified as **Low Value (L)**.